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Wireless magnetic sensors for traffic surveillance

Amine Haoui, Robert Kavaler, Pravin Varaiya *

Sensys Networks Inc., 2560 Ninth Street, Suite 211, Berkeley, CA 94710, USA

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Abstract

Sensys Networks' VDS240 vehicle detection system is a wireless sensor network composed of a collection of 3″ by 3″ by 2″ sensor nodes put in the center of a lane and a 6″ by 4″ by 4″ access point (AP) box placed 15′ high on the side of the road. A node measures changes in the earth's magnetic field induced by a vehicle, processes the measurements to detect the vehicle, and transfers the processed data via radio to the AP. The AP combines data from the nodes into information for the local controller or the Traffic Management Center (TMC). An AP communicates via radio directly with up to 96 nodes within a range of 150′; a Repeater extends the range to 1000′. This range makes it suitable to deploy VDS240 networks for traffic counts, stop-bar and advance detection, and measurement of queue lengths on ramps and at intersections, as well as parking guidance and enforcement. VDS240 is self-calibrating, IP-addressable and remotely monitored. Data are not lost because unacknowledged data packets are retransmitted. The accuracy of VDS240 for vehicle counts, speed and occupancy is comparable to that of well-tuned loops. Because the nodes report individual vehicle events, the AP also calculates individual vehicle lengths, speeds and inter-vehicle headways—measurements that can be used for new traffic applications. In July 2007, VDS240 systems were deployed in arterials and freeways in several cities and states, and 30 customer trials were underway in the US, Australia, Europe and South Africa.

Keywords: Traffic surveillance; Vehicle detection; Wireless sensors; Advanced traffic management systems

1. Introduction

Strategies for efficient road transportation management rely on a dense system of sensors that can quickly and accurately measure traffic. Adaptive signal control requires direct measurement or estimation of link free-flow speeds, queue discharge rates and intersection queue lengths (Mirchandani and Head, 2001). Freeway ramp control is much improved by algorithms that use ramp queue measurements (Sun and Horowitz, in press). An empirical analysis of California freeway data concludes that detectors should be placed every one-third of a mile to accurately measure congestion (Kwon et al., 2007). Solutions to the 'parking problem'

* Corresponding author. Tel.: +1 510 642 5270; fax: +1 510 642 7815.
E-mail addresses: haoui@sensysnetworks.com (A. Haoui), kavaler@sensysnetworks.com (R. Kavaler), varaiya@eecs.berkeley.edu (P. Varaiya).

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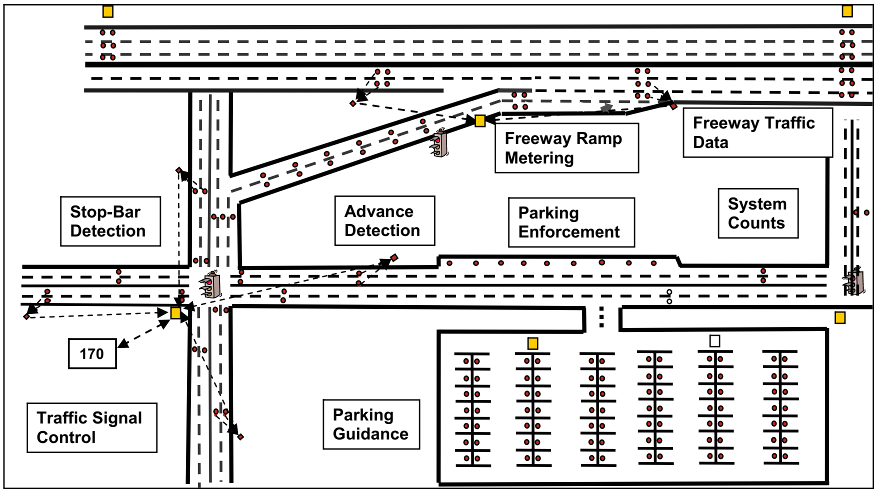

Fig. 1. The tiny circles denote vehicle detection sensor node locations for freeway traffic count stations, ramp metering, signal control, and parking guidance and enforcement. The small squares denote access points that communicate via radio with the sensor nodes.

in the US (Woodyard, 2006) and in Europe, where the search for parking constitutes 60% of the traffic in some cities (City Parking in Europe, 2006), depend on an accurate, real-time inventory of available parking places.

The tiny circles in Fig. 1 show the deployment of vehicle detection sensors for these applications. One conclusion is immediate from the figure. It would be impractical to use wires either to supply power to all the sensors or to communicate with them, because of their very high installation and maintenance costs. The only feasible approach is to use sensors that are battery-powered and communicate via radio. Wireless sensor networking is a breakthrough technology that combines sophisticated networking software with ultra low-power sensor and radio chips to enable sensing applications on a density and scale and at a cost that were unthinkable five years ago (Huang, 2003). This technology heralds a new era of traffic surveillance.

Sensys Networks, Inc. is a company that designs, manufactures and markets wireless magnetic sensor networks to accurately detect the presence and movement of vehicles in real time for the applications depicted in Fig. 1. Encouraged by the success of an initial prototype (Cheung et al., 2005), the co-authors founded Sensys in July 2003, and secured venture funding in mid-2004. In November 2005 Sensys began shipping systems for freeway count stations and arterial advance detection and count stations. Stop-bar presence detection systems were released in April 2007. In July 2007, VDS240 systems were deployed in streets and freeways in several cities and states, and 30 customer trials were underway in the US, Australia, Europe and South Africa.

Section 2 describes the VDS240 family of vehicle detector stations. Section 3 describes the basic signal processing algorithm for vehicle detection and summarizes experimental results comparing the performance of VDS240 with loops. Section 4 summarizes the case for the suitability of VDS240 for a pervasive traffic surveillance system.

2. VDS240 vehicle detection station

VDS240 is a local area wireless sensor network composed of a collection of sensor nodes (VSN240), an access point (AP240) and, possibly, a repeater (RP240) to extend the range of the access point; see Fig. 2. Data sheets for VDS240 and independently conducted product performance evaluations are available at www.sensysnetworks.com.

2.1. VSN240 vehicle sensor node

A node has electronic circuits that incorporate into a system a magneto-resistive sensor measuring the earth's magnetic field, a radio transceiver, an antenna, a microprocessor, memory and a battery with a 10 year

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Fig. 2. The VDS240 product family consists of sensor nodes (VSN240), access point (AP240) and a repeater (RP240).

lifetime. A node is packaged in a 3″ by 3″ by 2″ plastic case that is placed in the center of a lane in a 2 1/2″ deep cored hole, which is then filled by epoxy. The installation eliminates loop saw cuts and the circular holes produce the least amount of stress on the roadway. A node is installed in less than 10 min.

A node samples the three axes of the magnetic field at 128 Hz. A node operates in one of 6 modes: A. Raw-Z; B. Event; D. Raw-XYZ; E. Idle; F. Gated Raw-XYZ; and G. Parking. In mode A (respectively, D) the raw z axis (respectively, x, y, z axes) samples are sent to the AP. Mode F is like D, except that samples are sent only while a vehicle is present. The raw modes measure a vehicle's 'magnetic signature', which can be used for testing the node or to develop new applications, e.g., real-time vehicle classification and re-identification (Cheung, 2006).

Mode B is the normal mode. The microprocessor in the node processes the samples to determine when a vehicle arrives at, and departs from, the sensor node. The arrival and departure events of each vehicle are sent to the AP. All events are timestamped and synchronized across all nodes with a 1 ms resolution.

To determine vehicle arrival and departure events, the microprocessor uses a detection algorithm described in Section 3.1. The algorithm depends on several parameters. A database of vehicle signatures is used to select these parameters. The algorithm tracks the ambient magnetic field. Consequently, VSN240 is self-calibrating, eliminating the initial tuning and periodic re-calibration required by inductive loop, radar and video detectors. In order to calculate speed, two nodes are placed in the same lane at a fixed distance, creating a speed trap.

Each node has a unique network address included in every 50-byte data packet that it transmits to the AP. The AP acknowledges the data packets; unacknowledged packets may be retransmitted by the node. A node's operational mode can be changed at any time by the AP. The AP can also download new software to the node. The communication protocol between AP and a node is described in the Section 2.3.

The access point routinely measures and reports radio performance (received signal strength indicator or RSSI, packet loss, and link quality indicator or LQI) of each VSN240 in its local area. This permits remote diagnosis of the communications network. The performance of sensors and signal processing algorithms can be remotely monitored by using the raw mode. Lastly, VSN240 firmware can be updated 'over the air' to ensure that the installed base benefits from software feature enhancements and software bug fixes. This drastically reduces maintenance and upgrade costs.

2.2. AP240 access point

The AP is DC-powered and can support input voltages from 9 V to 24 V or from 36 V to 58 V allowing solar or line powering with battery backup, or standard POE (power over Ethernet) powering. Its host processor runs Linux, which allows sophisticated user applications to run directly on AP240. Its external communication is TCP-based, so that software development can take place elsewhere and ported to the host processor. Like VSN240, it is self-calibrating, requires no adjustment, and installs in minutes.

The AP is the 'base station' of the local area wireless network with up to 96 nodes. Its algorithms extract useful information from the data it receives from these nodes. For example, in freeway traffic applications, the 30-s count is just the number of vehicle arrivals at a node during 30 s; the 30-s occupancy is the fraction of vehicle 'on-time' (difference between vehicle departure and arrival times) during 30 s; a vehicle's speed is the distance between the two nodes in the speed trap divided by the difference between the vehicle arrival times; and a vehicle's 'magnetic' length is its on-time multiplied by its speed.

The measurement of individual vehicle arrival, departure and speed permit new applications. As one example, Coifman and Cassidy (2002) use sequences of individual vehicle lengths measured at two locations to estimate the travel time between those locations. As another application, Cheung et al. (2005) report an experiment in which the pattern of successive headways downstream of a traffic signal indicates whether the queue behind the signal is cleared during the green phase. Lastly, measurements of individual vehicle speed, platoon speed, and speed in adjacent lanes, can be combined with crash data to shed more light on the unsettled and important relationship between crash rate and speed (Kockelman and Ma, 2004).

The AP is enclosed in a 6″ by 6″ by 4″ box, and mounted between 12′ and 20′ high on a pole on the side of the road. It communicates via radio with nodes on the ground at a range of between 90′ and 175′. Using a RP240 repeater extends the communication range to 1000′. The radio transceivers in VSN240, RP240 and AP240 conform to the IEEE 802.15.4 PHY standard. The radio operates in one of sixteen 5 MHz channels in the 2.4–2.48 GHz band. Sensors and repeaters connected to the same AP operate on the same channel. Sensors connected to a repeater operate on a different channel. A repeater listens to the AP on the first channel and to its sensors on the second channel, and relays packets between the two channels.

2.3. Communication protocol

Communication in the local area network is governed by the Sensys Nanopower Protocol (SNP). SNP is a TDMA (time-division multiple-access) protocol, summarized in Fig. 3. Time is divided into 125 ms-long subframes, each subdivided into 64 time slots. Each sensor node is assigned a slot in which it can transmit 50 bytes of data. The figure shows slot 1 assigned to VSN A and slot 3 to VSN B. (A node that needs to transmit more data, for example in raw mode, is assigned more than one slot.)

Four slots are special. Slot 0 is used by the AP to synchronize the node clocks and to send global parameters for detection, transmit interval, etc. Slot 2 is used for software download. Slots 4 and 34 are used for acknowledgement of sensor data packets by the AP. If a node's data packet is not acknowledged, it retransmits the packet (up to a limit), which protects against data loss even when some transmitted packets are lost.

SNP's tight time synchronization permits the radio in the node to be in 'sleep' mode during slots when it is not transmitting or receiving. As a result the radio is awake less than 1% of the time, which enables the long battery lifetime. A detailed study (Coleri and Varaiya, 2006) explains why TDMA consumes a tiny fraction of the energy consumed by the more common CSMA (carrier sense multiple-access) protocol in multi-hop wireless sensor networks.

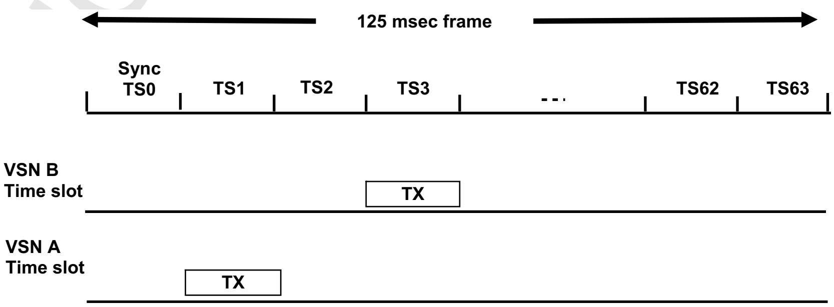

Fig. 3. VDS240 TDMA protocol for communication within the local area wireless network uses a 125 ms subframe divided into 64 slots. Each sensor node is assigned a fixed slot in which it can transmit 50 bytes of data.

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2.4. Local and wide-area interfaces

AP240 can activate contact closures to interface to standard 170, NEMA TS-1 and TS-2 and 2070 traffic controllers. For this purpose, Sensys provides standard form factor interface cards for connecting AP240 to the controller input file shelves. AP240 is connected to these interface cards via an RS-485 cable, which can also power the AP. This makes VDS240 backward 'loop compatible'. As illustrated in Fig. 4, a failed loop can be replaced by a sensor node, and the AP interface makes the node appear to the controller as a functioning loop. No change in the controller is needed. This makes the VDS240 a suitable loop retrofit system.

AP240 provides several other interfaces for data backhaul. A 10/100 Base T Ethernet interface accepts a DHCP or static IP address. (AP240 can be powered over the Ethernet.) Serial Port A interfaces to a cellular GPRS or $1 \times$ RTT CDMA modem or 115.2 kbps RS232C cable. Serial Port B interfaces to an optional GPS receiver, 115.2 kbps RS232C or RS485 cable.

AP240's TCP/IP support includes these protocols: telnet, ftp, http, ppp, pptp; tunneling to VPN allows connection to AP240 without a static address, and data can be encrypted. Acting as a SNP-TCP/IP gateway, AP240 supports TCP streams to applications, providing transparent access to sensor data. Multiple applications can ask for SNP packets based on packet type. The gateway can supply synchronization to applications. As noted earlier, third-party custom applications may be developed over IP (Ethernet or GPRS) and then ported to AP240. A 'C' API is available.

2.5. SNAPS

Large-scale deployment is assisted by the Sensys SNAPS server, which provides connectivity to APs over a wide area, data archiving, performance reports, and network element management. SNAPS computes counts, speed, occupancy, vehicle length and other statistics over various time intervals, as well as real-time individual vehicle speed, length and headway. Data can be pulled via telnet/ftp or pushed via the PeMS Interface. (PeMS (2006) is the California Freeway Performance Measurement System. PeMS archives freeway traffic data and computes numerous performance measures, including congestion and productivity.) On-board storage of data eliminates the problem of dropped data. Users may connect to SNAPS over the web; the TrafficDot applica-

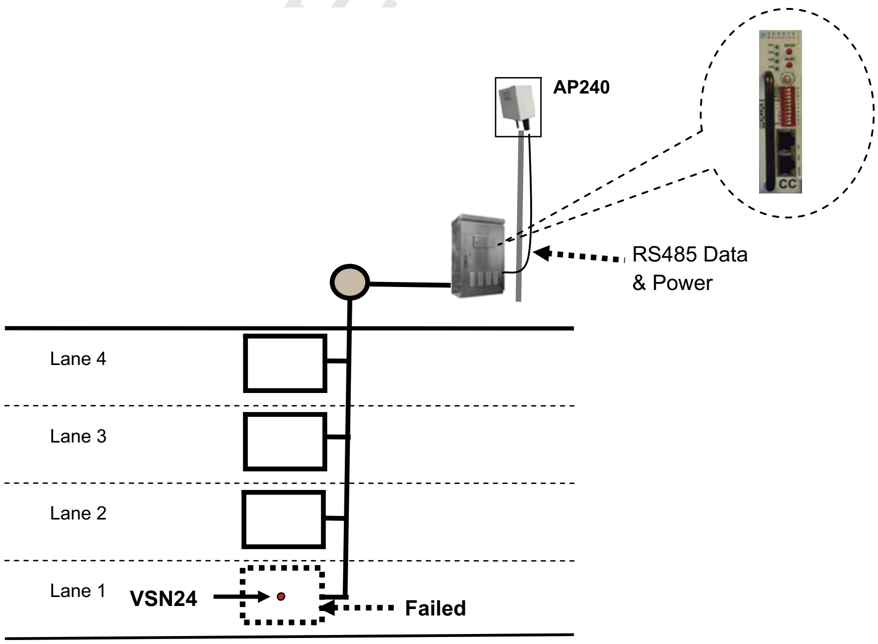

Fig. 4. VDS240 is 'loop compatible'. The failed loop in lane 1 is replaced by a sensor node and the AP240 interface provides the same input to the controller as a functioning loop.

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tion displays node events in real time. Fig. 5 depicts a deployment along a freeway encompassing several APs. Traffic data are brought via the internet to the customer desktop as soon as the VDS240 systems are installed.

3. Performance

The most fundamental decision made by a traffic surveillance system is vehicle detection: the decision whether a vehicle is present or not and, if it is present, the determination of the time instants of the vehicle's arrival and departure. The quality of all other measurements, such as counts, occupancy and speed, depends on the accuracy of vehicle detection.

3.1. Vehicle detection

Fig. 6 shows the vertical or z axis of the earth's magnetic field measured by a VSN240 sensor as a truck moves over the node. (This is the vehicle's z axis magnetic signature.) Because the vehicle causes a rapid change in the magnetic field, the vehicle can be detected by a threshold rule: a vehicle arrives when the magnetic field crosses a threshold, say α, from below and it departs when the field crosses a slightly lower threshold, say β, from above. (The earth's z axis magnetic field strength varies between 0.3 and 0.6 gauss over North America; the default threshold values recommended for VSN240 in North America are $\alpha = 40$ milligauss, $\beta = 23$ milligauss. These values indicate the large fluctuation in the magnetic field caused by a vehicle.) Two modifications are needed to make the threshold rule work in practice.

First, the ambient magnetic field drifts over time and space. Hence in order to keep fixed thresholds, the ambient field must be tracked and the measurements must be centered at the tracked value, as is the case with the plot in Fig. 6. Tracking is achieved by a first-order filter using sample values only when a vehicle is not present and when the magnetic field is not fluctuating significantly. Tracking the ambient field automates calibration and permits use of a fixed threshold. One consequence is that different copies of the VSN240 give the same measurements and inter-sensor node differences are virtually non-existent. By contrast, inductive loop measurements depend on how the loops were installed and how they are affected by pavement stress over time. Moreover, calibration of loop-based systems in the field is almost never done because it is prohibitively expensive, requiring expensive video processing to obtain ground truth. As a result, loop-based occupancy and speed measurements suffer from an unknown bias, making direct inter-loop comparison of these quantities meaningless (Jia et al., 2001).

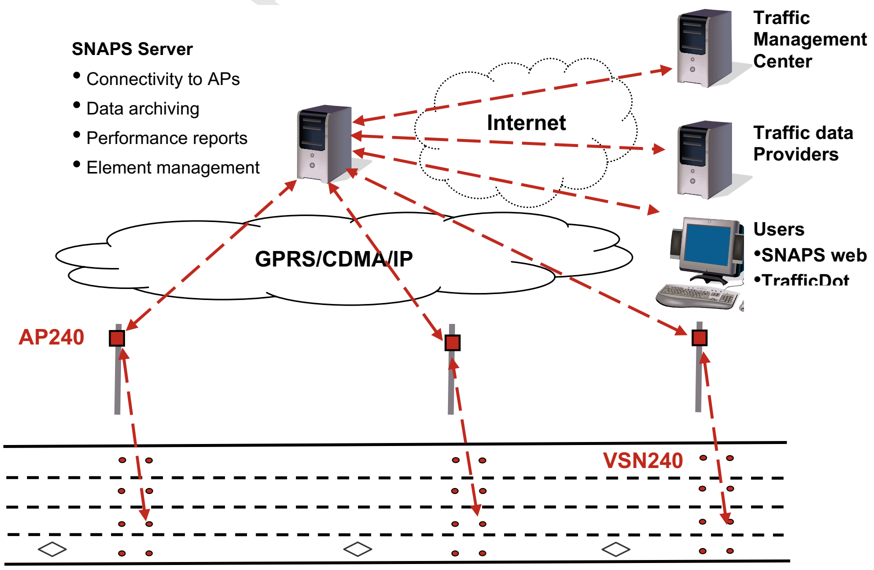

Fig. 5. The SNAPS server collects information from many AP's distributed over a wide area.

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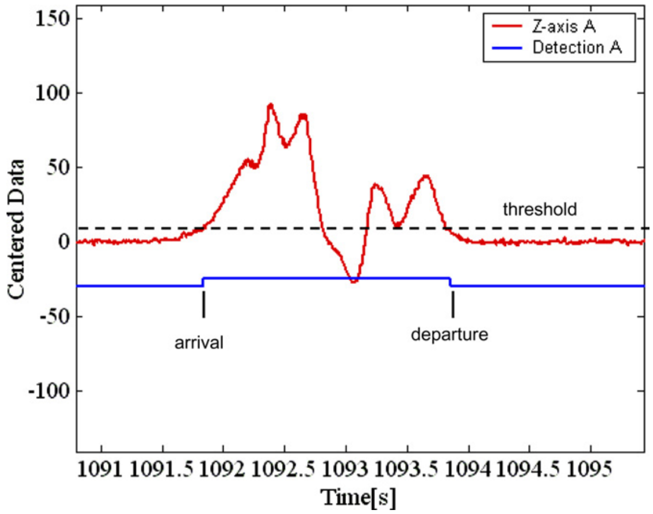

Fig. 6. Raw z axis measurement of earth's magnetic field as a truck moves over the sensor; threshold; and vehicle arrival and departure events.

Second, as seen in Fig. 6, the magnetic field may cross the threshold several times during the passage of the same vehicle, so a naïve implementation of the threshold rule will declare multiple vehicles in place of one vehicle. The figure also suggests the remedy: if a threshold up-crossing occurs very quickly after a down-crossing, it is likely that they are associated with the same vehicle. That is, if a down-crossing is followed by an up-crossing within a short time interval, ~~say~~ δ, the two 'spurious' crossings should be ignored. The correct choice of δ is a little tricky. Observe that the interval between the spurious crossings depends on the vehicle speed: if the vehicle is going at half the speed, the interval will be twice as long. Thus δ must be adapted (inversely) to the speed. Furthermore, as the magnetic signature depends on the spatial distribution of ferrous material in the vehicle, the number of spurious crossings and their location in the signature will vary. So the proper choice of δ (as well as α, β) is finally based on tests with a vehicle signature database. Lastly, the longitudinal or x axis magnetic field is also used in detection to reduce the spurious crossings.

A vehicle detection system makes two kinds of errors: it overcounts vehicles by mistaking more than one vehicle for a single vehicle; and it undercounts vehicles either by not detecting a vehicle or by 'merging' two vehicles into one. Experimental results assessing VD240 performance are given in Section 3.3.

3.2. Traffic statistics

Let $a(i)$ and $d(i)$ denote the arrival and departure times of the ith vehicle detected by a VSN240 sensor node. These times are sent to its AP240 access point. It is then straightforward to calculate the standard traffic statistics of 30-s counts and occupancy. The access point can also calculate the time headway $a(i) - d(i-1)$. In order to calculate speed in a particular lane, two nodes are placed at a distance D apart, and the access point estimates the speed of the ith vehicle as $v(i) = D/[a_{\rm d}(i) - a_{\rm u}(i)$. Here $a_{\rm u}(i)$ is the arrival of the vehicle in the upstream node and $a_{\rm d}(i)$ is the arrival at the downstream node. The 1magnetic' length of the ith vehicle is estimated as $l(i) = [d(i) - a(i)] \times v(i)$.

3.3. Experimental results

Sensys customers have commissioned several independent studies to determine the accuracy of VDS240 and other characteristics of the product, such as battery life, radio performance, effect of temperature, snow, rain, etc. The findings discussed below are from the study (Margulici et al., 2006) prepared for Caltrans by the California Center for Innovative Transportation (CCIT). Other studies are available at www.sensysnetworks.com.

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The objective of the CCIT study was to evaluate VDS240 as a possible replacement for inductive loops for the measurement of freeway traffic parameters. The study focused on data quality measures of timeliness, completeness, validity, and accuracy as defined in the (FHWA, 2004) report. The FHWA report recommends two additional measures, coverage and accessibility or usability, which are not meaningful in the context of the CCIT study.

Data from two inductive loop pairs in lanes 1 and 2 (the two lanes closest to the median) on I-80, near Berkeley, are compared with data from four VSN240 nodes placed in the center of the loops. The loops are maintained by the Berkeley Highway Lab (BHL), which is operated by CCIT. Loop outputs are sampled at 60 Hz. In addition to the loops, video footage served as a ground truth to evaluate count accuracy of the VDS240. Data were collected from 5 AM to 10 PM daily for two full weeks, during June and July, 2006.

3.3.1. Timeliness

This refers to the time lag between sensed events and the availability of the event data. Two time lags are relevant. The first is the delay between a VSN240 radio transmission of the data packet containing vehicle arrival and departure timestamps and the reception of the packet by the AP240 radio. The second is the delay between the access point's transmission of a data packet via the GPRS cellular system and its reception at the TMC which, in the study, is represented by the SNAPS server (see Fig. 5). This second time lag is independent of the Sensys VDS240. Of course, if the access point is connected via a serial port to a local controller, the second time lag would be absent.

Table 1 displays statistics of the time lag between the sensor node and access point: the mean packet delivery time is 0.13 s and 99% of packets are delivered within 0.40 s; the corresponding numbers for the GPRS delay between the access point and the SNAPS server are 1.00 s and 7.06 s.

3.3.2. Completeness

This refers to the availability to a TMC application of the event data generated at the source, and is measured by the ratio of the number of records received by the application to the expected number of records generated by the source (sensor node). Data incompleteness is caused by sensor failure, failure of the link between sensor node and access point or a GPRS outage between the latter and the SNAPS server. If there is such a failure no detection event will be reported. However, there may also be no detection event. In the absence of ground truth concerning the number of source events, the completeness ratio cannot be calculated, and an alternative test was constructed. It was determined that between 5AM and 10 PM on weekdays it was extremely likely that at least one vehicle would pass over each sensor within a 90-s 'activity interval'. Thus the test compared the fraction of 90-s activity intervals during which the VSN240 nodes and the associated loop detectors report events.

According to Table 5.2 in (Margulici et al., 2006), the nodes and loop detectors reported events in more than 99% of the 450,000+ activity intervals. This means that there was no 90-s interval during the two week experiment during which either the VSN240 node failed or data were lost by the node-to-access point link failure or by the GPRS link failure. Since we know that both radio links do fail, the absence of data loss implies that both the acknowledgement scheme in the SNP protocol over the VSN240-AP240 link and the TCP implementation over the GPRS link work well.

Table 1
Timeliness

Statistic	Time from node to access point (s)	Time from access point to SNAPS server (s)
Mean	0.13	1.21
Median	0.12	1.00
Std Dev	0.27	4.44
96 percentile	0.25	N/A
99 percentile	0.40	7.06

Source (Table 5.1 and Appendix B, Margulici et al. (2006)).

3.3.3. Validity

This refers to the plausibility of the reported measurements, and can be measured by the fraction of records that lie within a pre-specified expected range. Validity does not measure accuracy; rather it measures the fraction of measurements that are outliers. The CCIT study examines the validity of vehicle detection, speed, occupancy, volume (flow), and volume-occupancy pairs.

3.3.3.1. Vehicle detection. Let $d(i) - a(i)$ denote the *on-time* associated with the passage of a vehicle reported by a sensor node (see Fig. 6) or a loop detector. Selecting the minimum plausible on-time as 8/60 s and the maximum on-time as 10 s, leads to 95–100% valid VSN240 on-time records, as seen in Table 2. (Since the 'magnetic' on-time is shorter than the 'induction' on-time measured by a loop, the VSN240 is expected to report more on-times below 6/80 s.) *Vehicle speed* Taking a range of 3–99 mph as a valid speed estimate, the study finds that 99.8% of 25,000+ VSN240 30-s average speed estimates fall within this range, see (Margulici et al., 2006, Appendix B).

3.3.3.2. Valid occupancy. The valid range of 30-s average occupancy was taken as 0–47%, as recommended by FHWA, and confirmed by loop detector occupancy measurements, 99% of which were within this range. The study found that more than 99% of 25,000+ estimates made by each of the four VSN240 sensors were within this range (Margulici et al., 2006, Appendix B).

3.3.3.3. Valid volume. The study found that more than 99% of 25,000+ estimates of 30-s volume averages made by each of the four VSN240 sensors were within the range 0–2600 vehicles per hour per lane (Appendix B, Margulici et al., 2006).

3.3.3.4. Valid volume-occupancy. To determine the validity of the pair (volume, occupancy) 30-s estimates, the study constructs a region in volume-occupancy space that includes 99% of loop detector-based estimates, and found that between 95% and 97% of each of the four VSN240 estimates and 98% of the four loop-based estimates belonged to the region (Appendix B, Margulici et al., 2006).

3.3.3.5. Summary. The study concludes that VSN240 traffic measurements were in ranges considered valid 95% of the time—almost the same proportion as for the loops.

3.3.4. Accuracy

Estimating the accuracy of the VDS240 system or any traffic surveillance system is a multi-dimensional problem, because it is not possible to obtain ground truth for all measurements. This is obvious for occupancy and speed measurements and so all studies, including the CCIT study, take measurements provided by a well-calibrated loop-based system as ground truth. Unlike other studies, however, the CCIT study also tries to account for bias in the loop based 'ground truth', using measurements from matching loop pairs.

3.3.4.1. Counts. In principle, video recordings can provide ground truth for counts (but not speed or occupancy) and the study does that. However, even here one needs to be careful for two reasons. First, it is difficult to synchronize the independent clocks that generate the timestamps for the video frames, VDS240 and loop

Table 2
On-time validity

	Percent on-time above 6/80 s		Percent on-time below 10 s	
	VSN240	Loop	VSN240	Loop
Dn_Ln1	94.6	97.3	100.0	100.0
Up_Ln1	94.4	98.1	100.0	100.0
Dn_Ln2	95.3	99.0	100.0	99.9
Up_Ln2	95.5	98.9	99.9	99.9

Dn_Ln = Downstream lane, Up_Ln = Upstream lane. Source (Appendix B, Margulici et al. (2006)).

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events. To minimize the impact of synchronization error, the study compares video counts with VDS240 and loop counts over 5-min intervals. Second, when a vehicle is changing lanes, differences in vehicle counts are inevitable. (Because lane 1 is a high-occupancy vehicle or HOV lane, there are many lane changes.) Table 3 gives the ratio of (manual) video counts to each of four loop detector and VDS240 counts over six 5-min samples in light, medium and heavy traffic. Averaged across the six samples the ratio is 1.00 for loops and between 0.98 and 1.01 for VDS240.

3.3.4.2. Occupancy, volume, speed. The study compares 30-s average occupancy, volume, and speed calculated by loop detectors and the wireless sensors. The results are summarized in Table 4, reproduced from the study. The ‘range’ refers to the 10th and 90th percentile values of the 50,000+ 30-s average samples. Construction of the table requires matching the samples between loops (upstream and downstream in the same lane), between sensors, and between loops and sensors.

The correlation coefficient is between 0.96 and 0.99 between matched loop and sensor values, suggesting an almost perfect match between loop and sensor estimates of standard traffic statistics. The correlation coefficients are between 0.99 and 1.00 for 5-min averages, (Table 5.5, Margulici et al., 2006).

Table 4 also gives the bias or mean and standard deviation of the difference between matched samples. The numbers in the table are in units of the variables. The occupancy bias between loops and sensors is between 0.14% and 0.32%; the vehicle count bias is between 0.10 and 0.24 vehicles; and the speed bias is between 0.35 and 0.98 mph. Further, in the words of the report, “the bias between two wireless sensors in the same lane is almost non-existent [0.01 vehicle for counts and 0.07% for occupancy], suggesting that there are no noticeable differences due to manufacturing or calibration between two independent sensors [..] both the bias and standard deviation are less in the case of the wireless sensors self-comparison as they are for the loops self-comparison. This suggests that the wireless sensors produce more consistent results (Margulici et al., 2006, pp. 40–41).” ((Cofiman, 2004) compares radar detectors with loops at the same BHL test site and finds large lane-dependent bias and errors in radar count, occupancy and speed).

Table 3
Ratio of video count to sensor count over 5-min samples

	6–15% occupancy light–medium congestion				>15% occupancy heavy congestion				<5% occupancy light congestion				Average	
	Sample 1		Sample 2		Sample 3		Sample 4		Sample 5		Sample 6		All	
	LD	~~ws~~	LD	WS	LD	WS	LD	WS	LD	WS	LD	WS	LD	WS
Dn_Ln1	1.00	0.98	1.00	1.00	0.99	0.98	1.00	0.99	1.00	0.95	1.00	0.95	1.00	0.98
Up_Ln1	1.00	1.02	1.00	1.00	0.99	0.98	1.00	0.97	1.00	0.99	1.00	0.99	1.00	0.98
Dn Ln2	1.00	1.00	0.99	1.01	1.01	1.04	1.00	1.02	1.01	1.00	1.00	1.00	1.00	1.01
Up Ln2	1.00	1.00	1.02	1.01	1.07	1.04	1.04	1.02	1.00	1.00	1.00	1.01	1.00	1.01

LD = loop detector, WS = wireless sensor, Dn_Ln = Downstream lane, Up_Ln = Upstream lane. Source (Appendix B, Margulici et al. (2006)).

Table 4
Comparison of loop-based and VDS240 estimates of 30-s averages of occupancy, counts and speed

Variable	Range	Loop-Loop			Sensor–sensor			Loop–sensor		
		Bias	StdDev	cc	Bias	StdDev	CC	Bias	StdDev	CC
Occ (%)	1–24	0.06–0.28	1.04–1.30	1.00	0.06–0.07	0.72–1.13	1.00	0.14–0.32	1.10–1.74	0.98–0.99
Volume (#)	3–16	0.01–0.14	0.31–0.55	1.00	0.00–0.01	0.37–0.50	1.00	0.10–0.24	0.99–1.22	0.96–0.98
Speed (mph)	19–77	–	–	–	–	–		0.35–0.98	2.25–2.33	0.99

Bias = mean difference, StdDev = standard deviation of difference, and CC = correlation coefficient, of matched samples. Source (Table 5.6, Margulici et al. (2006)).

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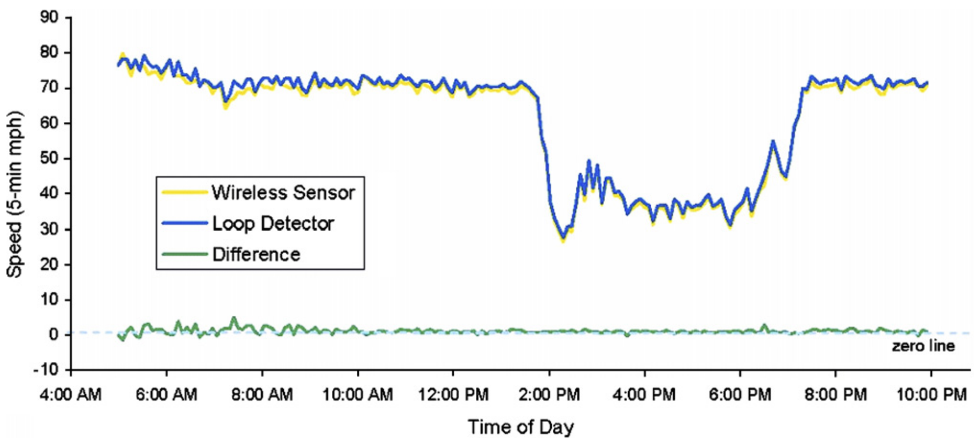

Fig. 7. 5-min average speed estimated by loop detector and VDS240 over one day, 5 AM–10 PM. Source (Fig. 5.3, Margulici et al., 2006).

Finally, Fig. 7, reproduced from the study, compares the 5-min average speed estimates over one day. The figure confirms the virtually perfect correlation in the measurements made by a well-calibrated loop-based system and VDS240.

3.3.5. Failures

As seen above, VDS240 systems installed directly 'out of the box' with no tuning or calibration, work with the accuracy of well-calibrated inductive loop detector systems. Typical loop-based systems in the field work very poorly by comparison. For example, 23% of 24,000 loops deployed in California freeways suffer large

Fig. 8. Installing a loop-based system (above) vs. VDS240 (below).

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data loss (PeMS, 2006); moreover the reasons for the loss are impossible to ascertain remotely. By contrast, the communication protocols and data buffering in VDS240 eliminate data loss; and the remote monitoring of the communication links can diagnose the cause of malfunction. Fig. 8 shows pictorially that it is much easier to install a VDS240 system than a loop-based system.

Vehicle detection systems fail and it is important to quickly and remotely detect the failure. Some loop failures, such as chattering, are quickly detected by statistical tests. But crosstalk—in which the detector in one lane reports the presence of vehicles in an adjacent lane—cannot be detected because the reported faulty measurements look normal. Similarly, misalignment, which causes radar detectors to incorrectly report measurements in an adjacent lane, is difficult to detect. Crosstalk and misalignment can occur because of the poor quality of installation or because of changes over time. VDS240 provides both qualitative and quantitative diagnostic information in the form of RSSI and LQI, both of which can readily be understood and adjusted very simply. As a result VDS240 never confuses traffic from one channel to another. At the same time, a sensor failure is quickly detected.

In summary, loop-based systems are failure-prone and inaccurate; moreover, their cost of installation and maintenance will grow over time because they require large labor time. Making new investments in obsolete loop technology is ill-advised. Radar-based systems may not provide the accuracy needed for applications like ramp metering; and they cannot be used for traffic signal control or stop-bar presence detection or queue measurements. They, too, require skilled labor for calibration and maintenance. The VDS240 requires no calibration, uses little labor time during installation, and is remotely maintained. Moreover, it serves all traffic surveillance applications depicted in Fig. 1. The lifecycle cost of a VDS240 system is a fraction of a loop-based system. Furthermore, most of the cost of VDS240 is due to electronics manufacture and assembly, both of which will decline with volume. This makes VDS240 the preferred choice for pervasive traffic surveillance.

4. Conclusion

It is a commonplace among professionals that effective implementation of standard ITS schemes of ramp metering and signal control can dramatically reduce congestion. For example, it is estimated that ramp metering could have eliminated 37% of the PM peak congestion in California freeways during October–December, 2005 (PeMS, 2006). Congestion reduction also provides large benefits from improved travel time reliability (Cambridge Systematics, 2001). Caltrans has recently begun to measure the impact of congestion by *lost productivity*—defined as the number of lane–mile–hours of capacity lost due to congestion. The lost productivity of California freeways during October 2006 is estimated at 14,000 lane–mile–hours per day (PeMS, 2006). Because of lack of data, we have no way of estimating congestion loss on streets.

Despite the consensus about the efficacy of ITS schemes, they are rarely implemented, even in a progressive state like California. More sophisticated schemes for co-ordinated ramp and signal control, rapid transit, traveler information, incident detection and management, parking guidance and demand management, remain confined to research journals with no plans for real-world testing. One reason for this poor implementation record is the lack of adequate detection, in terms of coverage and accuracy. The quality of loop detector data is widely perceived as being so poor that engineers often do not trust the data.

Thus implementation of ITS schemes requires a pervasive system of accurate traffic sensors. With its high cost, inadequate accuracy and difficulty of deployment, loop detector technology—the loops, detector cards, wiring for power and communication, and controllers—is incapable of serving this function. Wireless sensor network technology appears ideally suited. It requires no wires, so that it can be deployed wherever necessary. It is as accurate as well-calibrated loops. It provides individual vehicle measurements, permitting the development of new applications.

VDS240's low cost, ease of installation and maintenance, flexibility of deployment, accuracy, IP networking and data integrity make it suitable for pervasive traffic surveillance. VDS240 is backward 'loop compatible': a defective loop, when replaced by a node, appears to the controller like a functioning loop. VDS240 is 'future proof'. First, as new applications are developed, the software can be downloaded to existing systems. In the future, road system surveillance will include other sensing modalities, such as temperature, pressure, humidity, light, and pollutants, for which low-power, miniature sensors are becoming available. These sensors can be deployed in nodes alongside the magnetic sensor nodes described here, while communicating with the same

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access points. Alternatively, the sensors could be incorporated within an enhanced version of VSN240. Lastly, the addition of a 802.1 1~~x~~ transceiver to the AP240 can make it serve as a gateway or as a node in the future vehicle-infrastructure integration (VII) network, in which data can be collected and transferred between vehicles and the roadside (Misener, 2005).

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References

Cambridge Systematics, Inc, 2001. Twin Cities Ramp Meter Evaluation. Final Report.

Cheung, S.-Y., Coleri, S., Dundar, B., Ganesh, S., Tan, C.-W., Varaiya, P., 2005. Traffic measurement and vehicle classification with a single magnetic sensor. Transportation Research Record (1917), 173–181.

Cheung, S.-Y., 2006. Traffic Surveillance by Wireless Sensor Networks. Ph.D dissertation, Department of Mechanical Engineering, University of California, Berkeley, CA.

City Parking in Europe. <http://www.city-parking-in-europe.net/> (accessed 12.26.06).

Coifman, B., Cassidy, M., 2002. Vehicle reidentification and travel time measurement on congested freeways. Transportation Research: Part A 36 (10), 899–917.

Cofiman,B., 2004. An Assessment of Loop Detector and RTMS Performance. California PATH Research Report UCB-ITS-PRR-2004-30. Online at <http://www.path.berkeley.edu/PATH/Publications/>.

Coleri, S., Varaiya, P., 2006. PEDAMACS: Power efficient and delay aware medium access protocol for sensor networks. IEEE Transactions on Mobile Computing 5 (7), 920–930.

Federal Highway Administration, 2004. Traffic Data Quality Measurement: Final Report. Online: <http://knowledge.fhwa.dot.gov/cops/hcx.nsf/All+Documents/DEDC37CA356402F185256F9B00595E89/$FILE/Traffic%20Data%20Quality%20Measurement%20-Final.pdf> (accessed 02.08.07).

Huang, G.T., 2003. Casting the Wireless Sensor Net. MIT Technology Review. Online: <http://www.technologyreview.com/InfoTech/13235/> (accessed 12.26.2006).

Jia, Z., Chen, C., Coifman, B., Varaiya, P., 2001. The PeMS algorithm for accurate, real-time estimates of *g*-factors and speeds from single loop detectors. In: Proceedings 2001 IEEE Intelligent Transportation Systems Conference, Oakland, pp. 527–532.

Kockelman, K., Ma, J., 2004. Freeway speeds and speed variations preceding crashes, within and across lanes. In: Proceedings of 83rd Transportation Research Board Annual Meeting, Washington, DC.

Kwon, J., Petty, K., Varaiya, P., 2007. Probe vehicle runs or loop detectors? Effect of detector spacing and sample size on the accuracy of freeway congestion monitoring. In: Proceedings of 86th Transportation Research Board Annual Meeting, Washington, DC.

Margulici, J.D., Yang, S., Tan, C.-W., Grover, P., Markarian, A., 2006. Evaluation of Wireless Traffic Sensors by Sensys Networks: Final Report. California Center for Innovative Transportation. Online: <http://www.calccit.org/resources/2006%20PDF/CCIT%20Task%20Order%205%20-%20Final%20Report.pdf> (accessed 12.28.06).

Misener, J., 2005. Vehicle-Infrastructure Integration (VII) and safety: rubber and radio meet the road in California. Intellimotion 11 (2), 1–3, Online: <http://www.path.berkeley.edu/PATH/Intellimotion/IM11-3_high.pdf> (accessed 01.02.07)..

Mirchandani, P., Head, L., 2001. A real-time traffic signal control system: architecture, algorithms and analysis. Transportation Research Part C 9, 415–432.

PeMS. California Freeway Performance Measurement System. <http://pems.eecs.berkeley.edu> (accessed 12.28.06).

Sun, X., Horowitz, R., in press. A set of new traffic-responsive ramp-metering algorithms and microscopic simulation results. In: Proceedings of 85th Transportation Research Board Annual Meeting, Washington, DC. Transportation Research Record. Q1

Woodyard, C, 2006. Need a parking place? Good luck. USA Today, 12/11/2006. Online: <http://www.usatoday.com/money/autos/2006-12-11-parking-usat_x.htm> (accessed 12.26.06).

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